

Databases

Module 5 — AWS Technical Essentials

🛠️ Managed vs Unmanaged

Unmanaged (DB on EC2)

- You manage everything
- OS patches, backups, HA
- Full control, full burden

Managed (RDS, DynamoDB)

- AWS handles operations
- Auto backups, patching, HA
- Focus on data and queries

🗃️ Amazon RDS & Aurora

Relational Database Service — managed SQL databases in the cloud.

- MySQL, PostgreSQL, MariaDB, Oracle, SQL Server
- Automated backups (up to 35 days)
- Push-button Multi-AZ failover
- Read replicas for scaling reads
- Encryption at rest and in transit

> **Amazon Aurora**
>
> AWS-built, MySQL/PostgreSQL compatible. 5x MySQL throughput. Storage auto-scales to 128 TB. 6-way replication across 3 AZs.

🎯 Purpose-Built Databases

AWS philosophy: use the right database for each workload.

- **DynamoDB:** key-value / document (NoSQL)
- **ElastiCache:** in-memory caching (Redis, Memcached)
- **Neptune:** graph relationships
- **Redshift:** data warehousing / analytics

⚡ Amazon DynamoDB

Fully managed NoSQL — key-value and document database at any scale.

- Single-digit millisecond latency
- Serverless: no instances to manage
- Automatic scaling and replication
- Built-in encryption and backups

🔑 DynamoDB Key Concepts

- **Table:** collection of items (like rows)
- **Partition Key:** unique identifier for each item
- **Sort Key:** optional, enables range queries
- **Capacity:** on-demand or provisioned mode

🎯 Choosing a Database

Use RDS/Aurora when...

- Structured data with relationships
- Complex SQL queries needed
- ACID transactions required

Use DynamoDB when...

- Key-value access patterns
- Massive scale, low latency
- Schema flexibility needed

🛠️ Lab Time

Create a DynamoDB Table

⏱️ 15 minutes

- Create a table for employee records
- Add items and query by partition key
- Connect the table to the app